The first-return map is an exact symplectomorphism and the boundary first-return time is a boundary integral

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Editorial note. The counted statement is the author's labelled lemma 'l:return_time_map' (source lines 1462-1472); the proof environment begins on the next line after \end{lem} apart from one blank line, so the pairing is by adjacency.

Editorial note. Two context spans are retained: the author's equation defining the normalised one-form and the author's almost-Besse normalisation lemma whose first part the proof invokes; every label used inside the retained text is therefore defined inside the package.

Editorial note. The retained preamble is a minimal editorial adapter supplying only the plain text abbreviations that the publisher class used to define, followed by the native author macro block of the pinned source.

Editorial note. No retained line refers to a label, citation or figure outside the retained spans, so the package declares no external reference.

Editorial scope. One original result of the pinned publisher article, counted once: the first-return map and boundary return-time formula (author label l:return_time_map).

Allowed standard prerequisites (editorial).

- First-return map and time. The first-return map phi and the first-return time tau are defined on the cross-section of the Reeb flow of the normalised form.. Both are the objects of the statement and of the computation.
- Exact symplectomorphism. A symplectomorphism is exact when the pull-back of the primitive one-form differs from it by an exact form.. The first conclusion is that phi^* nu = nu + d tau.
- Normalised form and its boundary behaviour. tilde-lambda_0 = dt + pr_1^* nu_0 on the interior, and its restriction to the boundary agrees with nu_0.. They are the two ingredients used at source lines 1486-1492 to evaluate the Reeb integral along the orbit segment.

$$\tilde{\lambda}_0 := \tilde{\imath}^* \lambda_0 = dt + \mathrm{pr}_1^* \nu_0 \tag{1}$$

Lemma 0.1. *i. The pull-back of ν via the inclusion $\partial\Sigma \hookrightarrow \Sigma$ satisfies*

$$\nu|_{\partial\Sigma} = \nu_0|_{\partial\Sigma}.$$

ii. The 2-form ω vanishes at all points of $\partial\Sigma$, i.e.

$$\omega_z = 0, \qquad \forall z \in \partial\Sigma.$$

iii. The Reeb vector field $R_{\tilde{\lambda}}$, a priori only defined on the interior of $\Sigma \times S^1$, admits a smooth extension to $\Sigma \times S^1$ that is tangent to the boundary $\partial\Sigma \times S^1$.

iv. For all $\varepsilon > 0$ there exists $\delta > 0$ such that, if the above contact form λ further satisfies $\|R_\lambda - R_{\lambda_0}\|_{C^2} < \delta$, then $\|R_{\tilde{\lambda}} - \partial_t\|_{C^1} < \varepsilon$. Here, the norms are associated with arbitrary fixed Riemannian metrics.

Lemma 0.2. *The first-return map ϕ is an exact symplectomorphism of (Σ, ω), and more precisely*

$$\phi^*\nu = \nu + d\tau.$$

The boundary restriction of the first return time τ is given by

$$\tau(z) = 1 + \int_{\{s \mapsto \phi_s(z)\}} \nu, \qquad \forall z \in \partial\Sigma.$$

Proof. The first statement follows by the well-known computation

$$\phi^*\nu = (\phi_{\tilde{\lambda}}^{\tau(z)})^*\tilde{\lambda}|_{\Sigma\times\{0\}} + \phi^*(\tilde{\lambda}(R_{\tilde{\lambda}}))d\tau = \nu + d\tau.$$

For each $z \in \partial\Sigma$, the curve

$$\zeta_z : [0,1] \to \partial\Sigma \times S^1, \qquad \zeta_z(s) := (\phi_s(z), s) = \phi_{\tilde{\lambda}}^{\tau_s(z)}(z, 0),$$

is a reparametrization of the restriction of the orbit of $(z, 0)$ by the Reeb flow of $\tilde{\lambda}$ to the interval $[0, \tau(z)]$, which makes one full turn around the second factor of $\partial\Sigma \times S^1$. Therefore,

$$\tau(z) = \int_{\zeta_z} \tilde{\lambda} = \int_{\zeta_z} \tilde{\lambda}_0 = \int_{\zeta_z} \big(ds + \mathrm{pr}_1^*\nu\big) = \int_{\zeta_z} ds + \int_{\mathrm{pr}_1\circ\zeta_z} \nu = 1 + \int_{\{s\mapsto\phi_s(z)\}} \nu,$$

where the second equality follows by Lemma 0.1(i), and the third one from (1). □

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